

# Introduction to Artificial Intelligence

Week 3

# Agenda

- What is an Agent
- PEAS
- Properties of the Environment
- Types of Agents

# Agents

Mr. Anderson….



# What is an Agent?



- An agent is anything which:
  - Lives in an environment
  - Can perceive the environment via its sensors
  - Can act upon the environment via its actuators
- Agents also have memory of the current and past perceptions of the environment
  - Percept – the current state
  - Percept Sequence – the complete history of Percepts

# Humans as Agents



- Environments
  - Home
  - School
  - Work
- Sensors
  - Eyes
  - Ears
  - Mouth
  - Nose
  - Skin
- Actuators
  - Hands
  - Feet
  - Speech

# Rationality



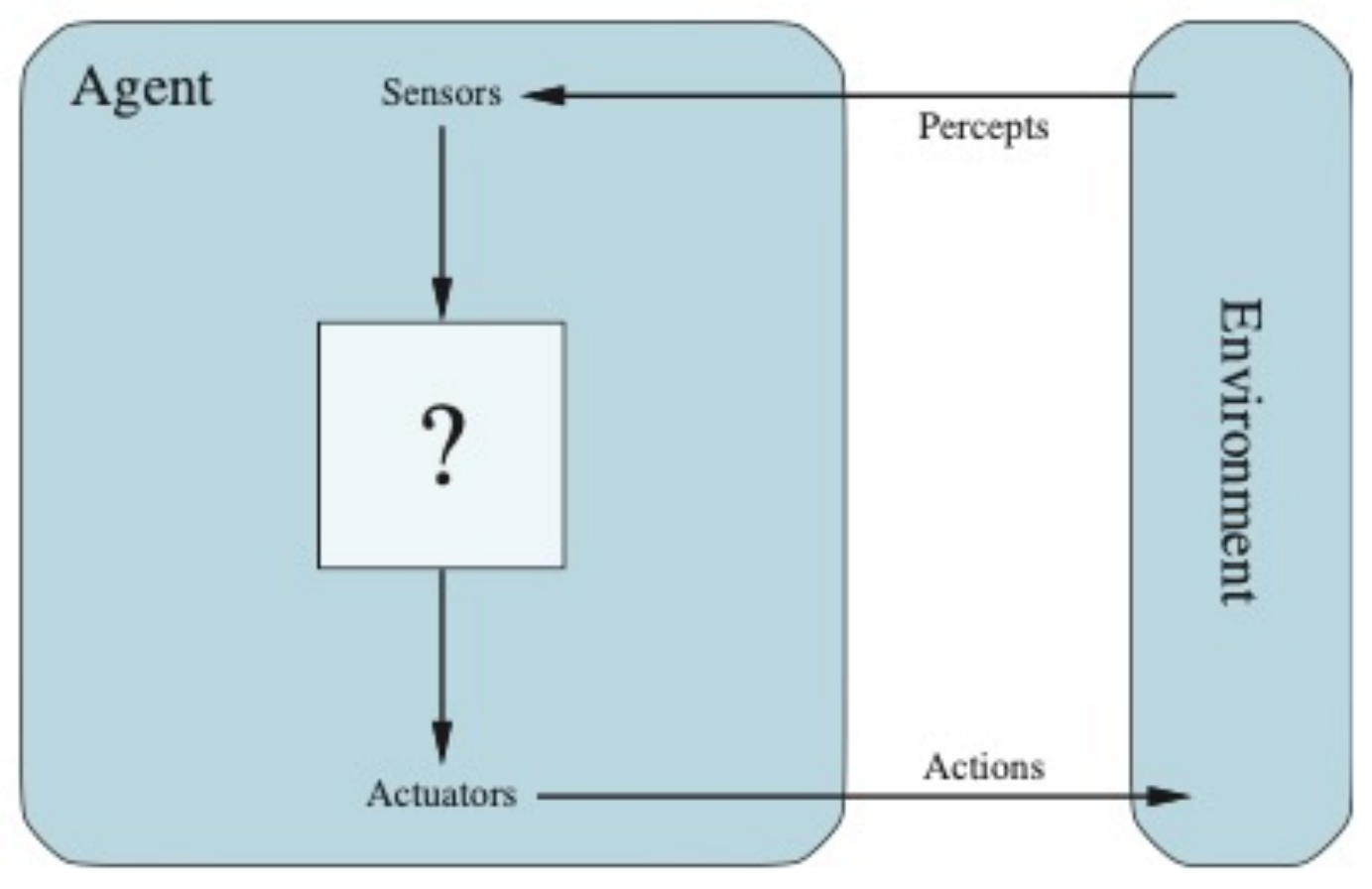

- What is rational at any given time depends upon:
  - The performance measure defines success
  - The agent's prior knowledge of the environment
  - The actions which an agent can perform
  - The agent's percept sequence to date
- Definition:
  - For Each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has

# Agents are Rational, Not Necessarily Omniscient



- A jester knocks over the king's wine
  - "What an idiot!"
  - Did the jester make a cartwheel and miss the landing?
  - Did the jester see it being poisoned and the king did not?
- Agents need to gather information before making a decision of an action
  - "Why did you spill my wine jester?!"
- Note that agents may perceive differently from each other
- Agents have a degree of autonomy
  - The King does not control every action of the jester

# PEAS – Performance Measure, Environment, Actuators, Sensors





# PEAS – Performance Measure, Environment, Actuators, Sensors



- Agent Type
  - Jester
- Performance measure
  - The King is Happy
- Environment
  - The Court
- Actuators
  - Tumbling
  - Juggling
  - Joke Telling
- Sensors
  - Hearing Laughter
  - Seeing Smiles

# Properties of the Environment (1)

- Fully Observable v. Partially Observable
- Single v. Multiple Agent
- Deterministic v. Stochastic
- Episodic v. Sequential
- Static v. Dynamic
- Discrete v. Continuous
- Known v. Unknown

# Properties of the Environment (2)



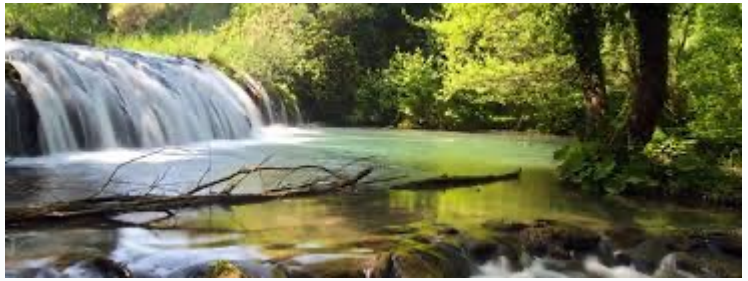
- Fully Observable v. Partially Observable
  - Can the agent see everything in the world via its sensors?
  - Does it miss elements which are relevant to the decision making process?
  - Unobservable for agent without sensors
- Single-agent v. Multiagent
  - Is there only one agent within the space?
  - Does there need to be some form of *competitive* or *cooperation* between agents?
  - How do the agents communicate?

# Properties of the Environment (3)



- Deterministic v. Stochastic
  - Can we determine the next state of the environment given the previous state and the action we are applying into it?
  - We can be certain in environments which are fully observable and deterministic
- Episodic v. Sequential
  - Episodic environments allow for atomic actions not dependant upon those previous
  - Sequential actions are those which the current decision has consequences on future actions

# Properties of the Environment (4)



- Static v. Dynamic
  - Does the environment change as the agent is thinking about its actions?
  - Semi-dynamic if the environment does not change as the time passes by but the agent's performance score does
- Discrete v. Continuous
  - Are there states in the environment, or does the environment smoothly act over time
    - Chess – discrete movements
    - Jenga – continuous actions of physics on the blocks
- Known v. Unknown
  - Does the designer of the agent have full knowledge of the rules of the environment?

# The Worst Case



- An environment which is partially observable, multiagent, stochastic, sequential, dynamic, continuous, and unknown
- Ordering in a new restaurant in a new country
  - Can't see how they make the food
  - Have to deal with the waiter, other customers
  - Can't predict when the food will come or how exactly it will taste
  - Depending on what I order, I might order more or might order an ambulance
  - Maybe they don't have anymore what I would like to order
  - Time keeps on ticking
  - Don't know the cultural rules



# Testing an AI



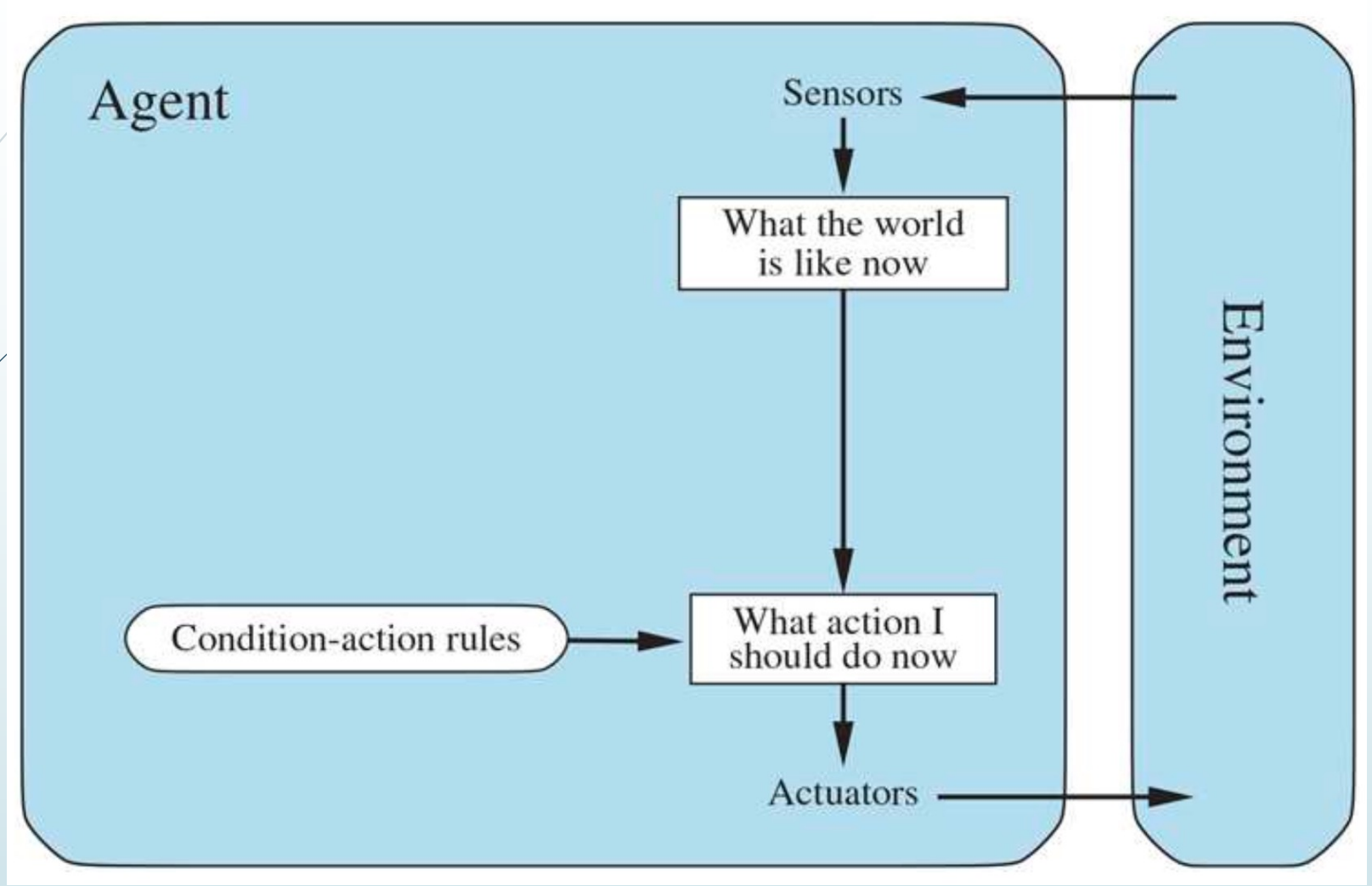
- We do not just want one environment but an environment class
  - Run multiple simulations with many different changes to the environment
- Self-driving Car
  - Traffic level
  - Lights
  - Weather
- Collectable Card Game Player
  - Against many other decks
  - Other player agents



# Types of Agents

1. Simple reflex agents
2. Model-based reflex agents
3. Goal-based agents
4. Utility-based agents

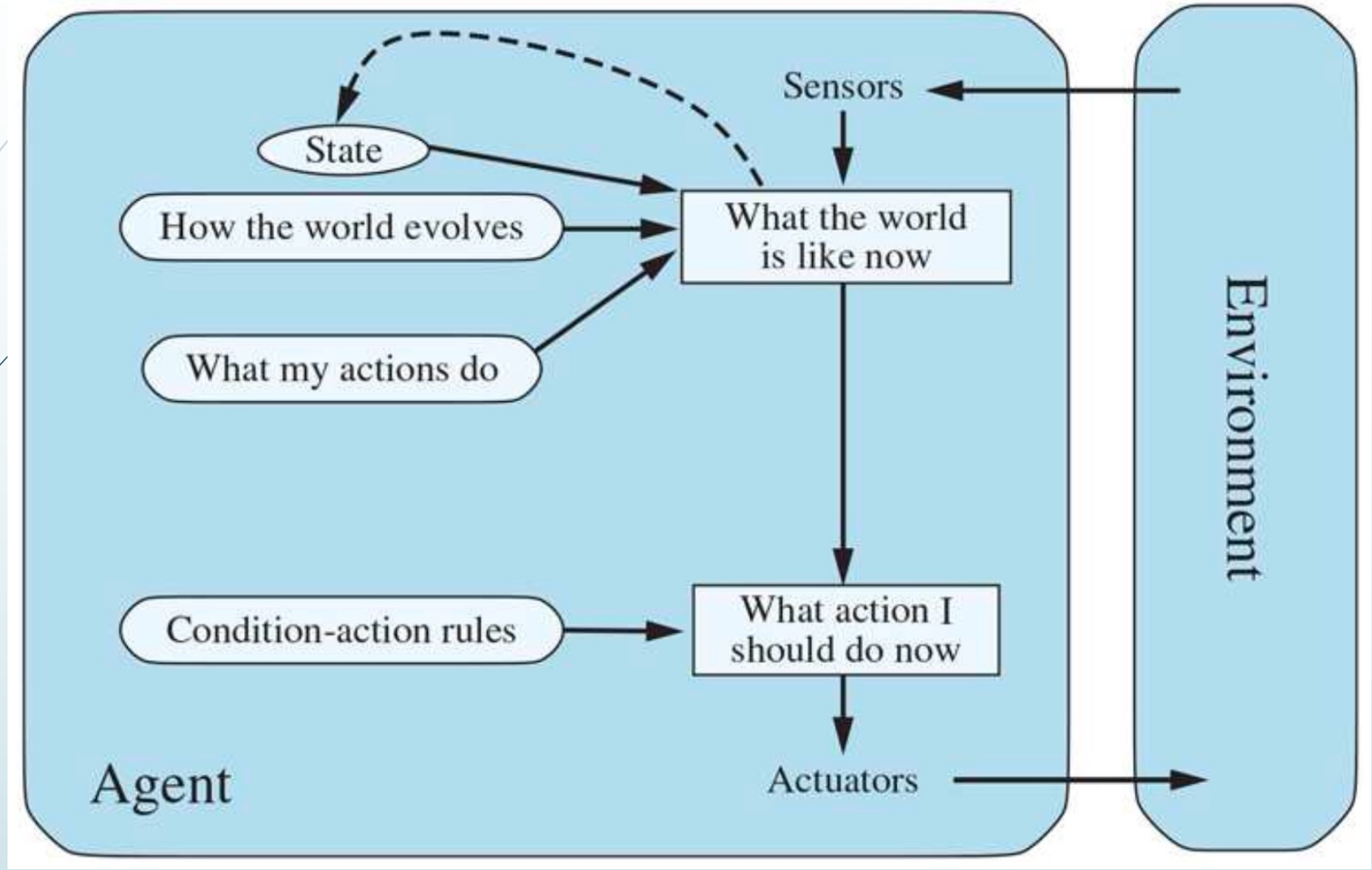
# Simple Reflex Agents



# Simple Reflex Agents

- Only current percept matters
- Work only in fully observable environment
- Too many condition-action rules may be required

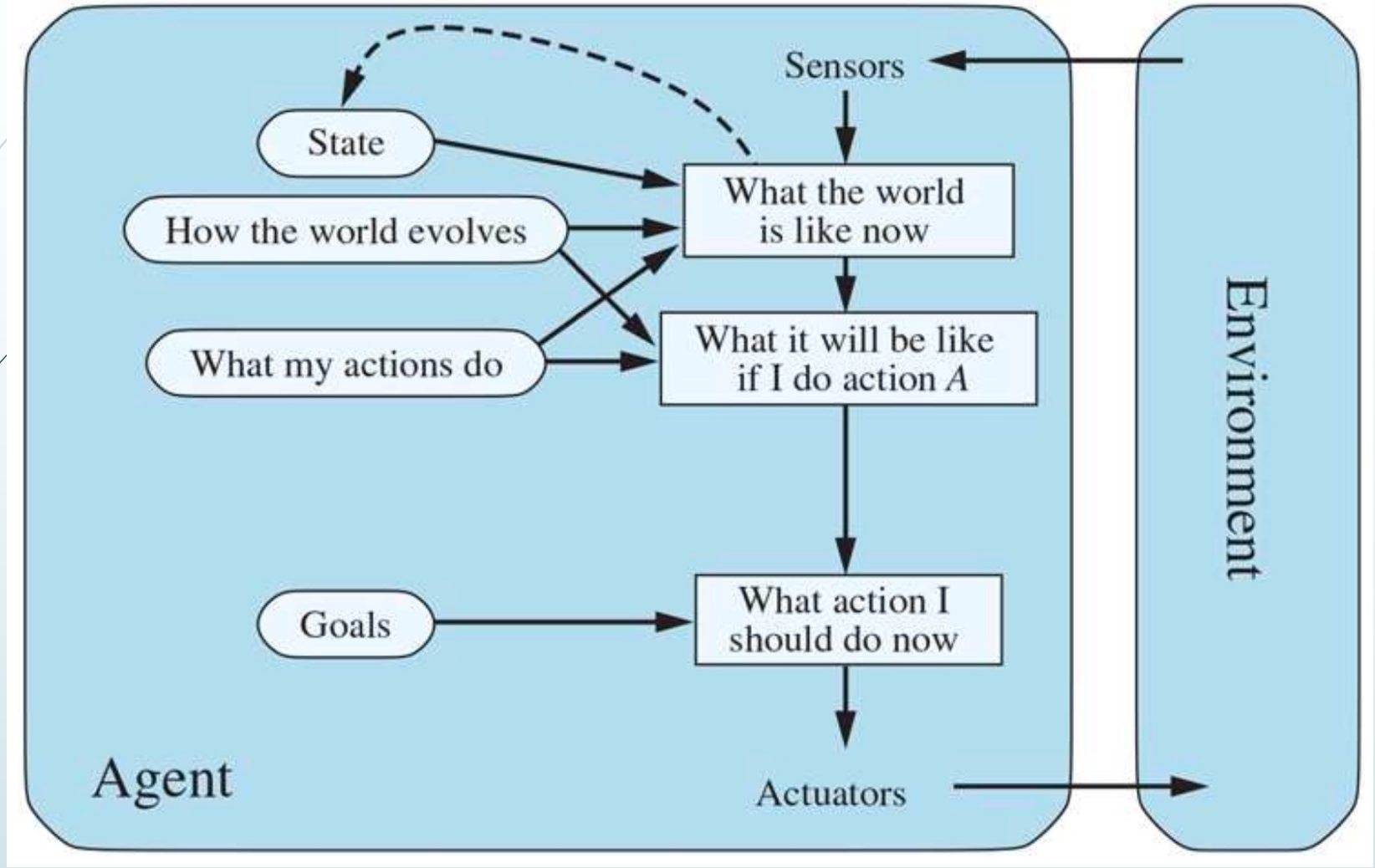
# Model-based Reflex Agents



# Model-based Reflex Agents

- Work better in partially observable environment because of keeping track of the world
- Internal state exists
- In partially observable environment not always possible to answer “What the world is like now?”. Sometimes agent tries to guess
- Still, too many condition-action rules may be required
- Not flexible enough

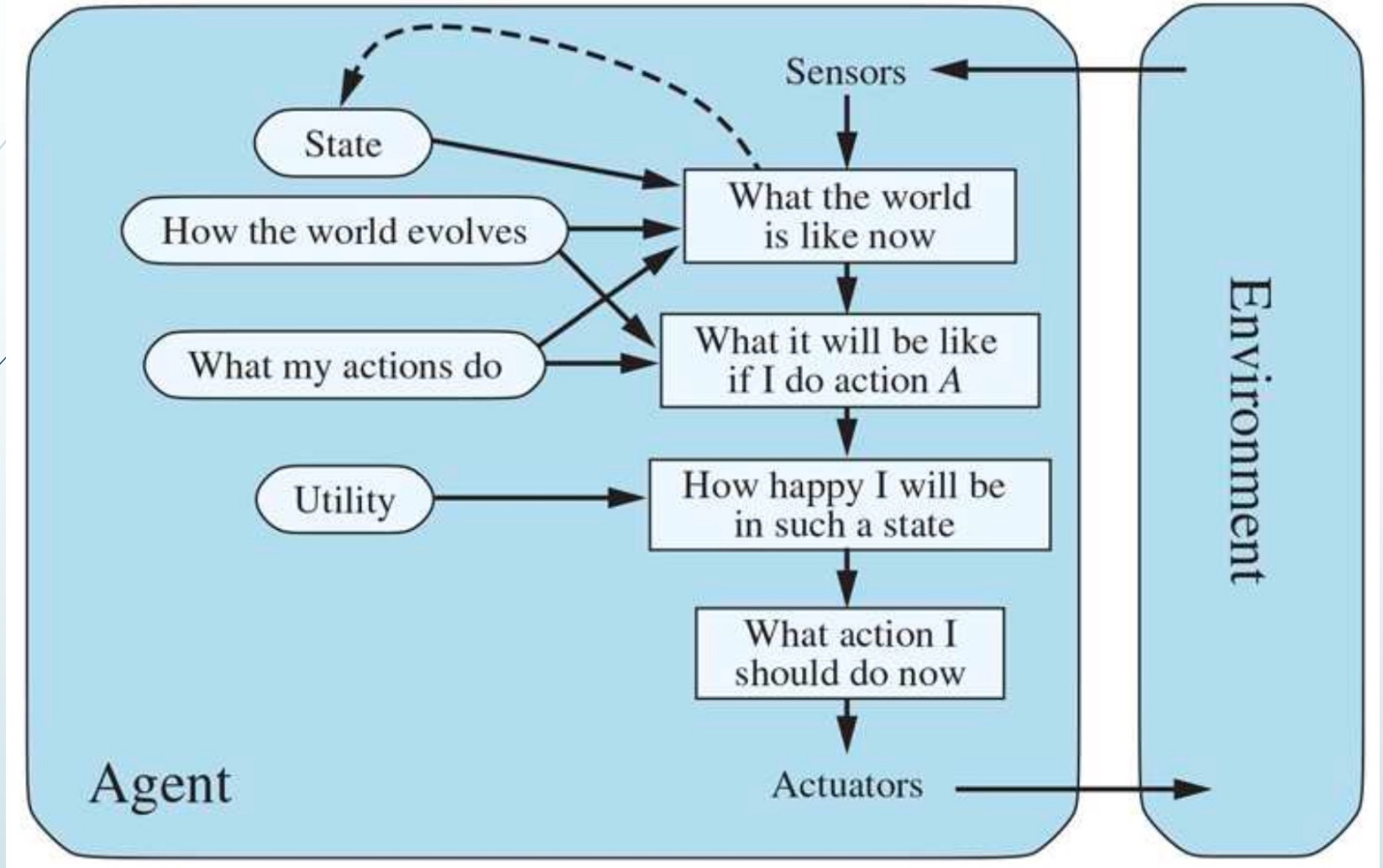
# Goal-based Agents



# Goal-based Agents

- More flexible than previous agent types
- Goal replacement does not entail rewriting of all the rules
- May be hard to choose between multiple goals

# Utility-based Agents



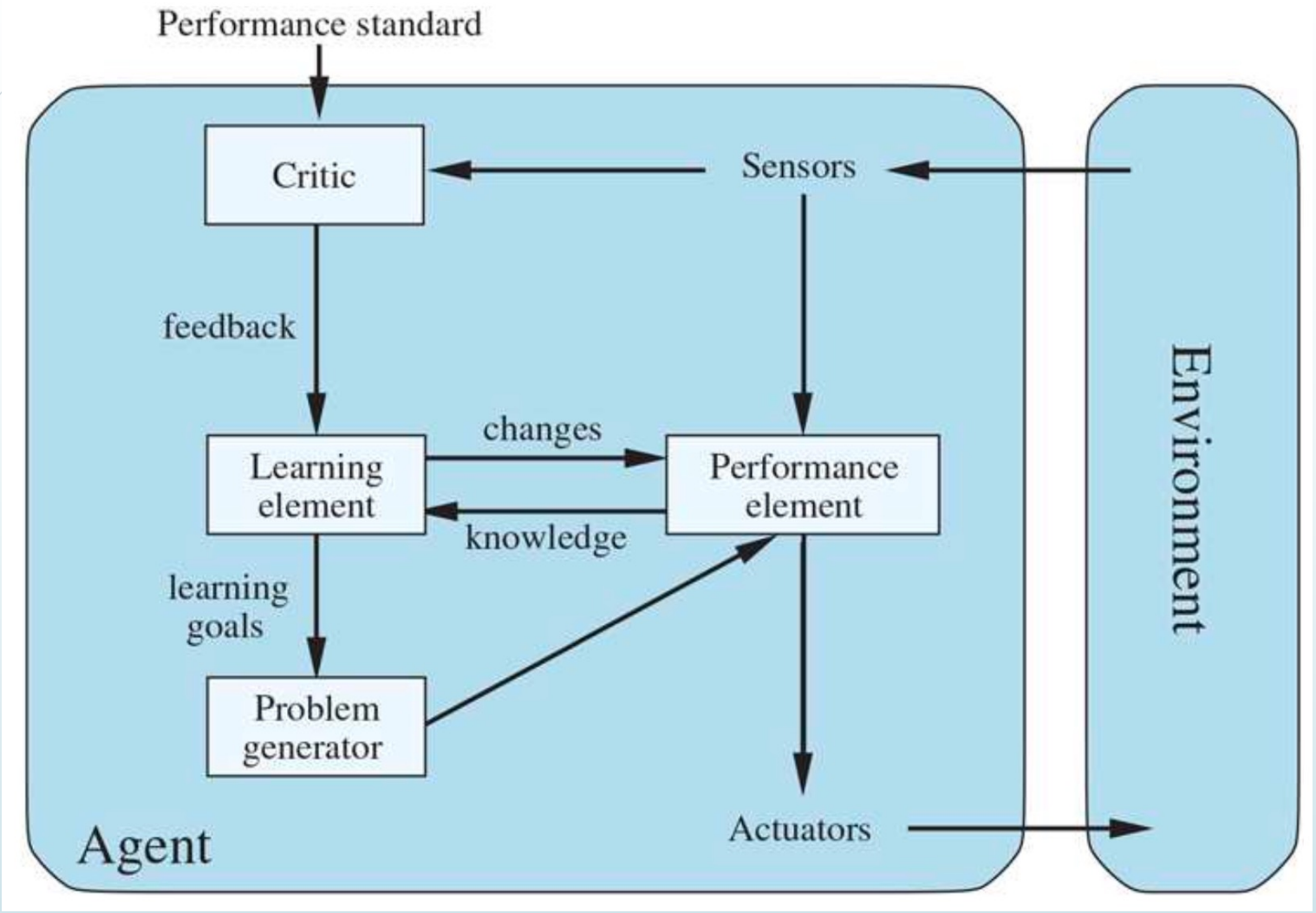
# Utility-based Agents

- Allow to maximize the "happiness" of the agent
- Allows to find the tradeoff between the conflicting goals
- More difficult to implement

# Learning Agents

- In 1950 Alan Turing proposed to build learning machines (agents) and then to teach them
- Any type of agent (model-based, goal-based, utility-based, etc.) can be built as a learning agent (or not)
- **Performance element** is what we previously considered as entire agent
- **Leaning element** is responsible for making improvements
- The learning element receives a feedback from the **Critic**
- **Problem generator** is responsible for suggesting actions that will lead to new and informative experiences

# Learning Agents



# Summary

- What is an Agent
- PEAS
- Properties of the Environment
- Types of Agents

Next week: Expert Systems